

Dual ordinal spectral ranking of journals and institutions

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March 7, 2026

Abstract

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval. Institutions fund individual researchers, implement training, oversee compliance, and maintain physical and information infrastructure. We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from journal–institution citation flows. A single coupling parameter χ apportions reference attention between source and institution channels, nesting conventional single-mode rankings as special cases. We apply the method to the economics and commerce literature using OpenAlex data and report rankings and their sensitivity to model parameters.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

Overview of problem: journal and institution influence ranking. [Placeholder — to be drafted.]

Previous work on journal ranking. [Placeholder — to be drafted.] Vigna [2016] provides an overview of spectral ranking methods, highlighting key methodologies and their sometimes unrecognised commonalities.

How institutions are ranked. Institutional rankings, dominated by QS, Times Higher Education (THE), and ShanghaiRanking (ARWU), blend bibliometrics with reputational surveys, teaching metrics, and internationalisation. QS 2026 ranks MIT #1, emphasising academic reputation (30%) and citations per faculty (20%); THE prioritises research income and industry links; ARWU focuses on top-journal publications and Nobel/Fields prize winners.

The Leiden Ranking counters with field-normalised bibliometrics (MNC, top 10% publications), revealing niche leaders but lacks QS/THE’s visibility due to the absence of reputation scores. Spectral methods remain rare at the institutional level; the dual ranking developed here exploits this gap, offering recursive prestige within feasible disciplinary bounds.

Social system of citations and actor-network theory. In citation analysis, distinguishing the mathematical structure of networks from the motivations of referencing proves analytically fruitful. Dual spectral rankings of journals and institutions—derived from Katz or Pinski–Narin iterations on a bipartite citation matrix—treat citations as structural relations that generate influence scores without presupposing authorial intent. Here we draw on the Social System of Science Citation Theory (SSCT), which, building from Luhmann via Lehmann, conceptualises citations as autopoietic communications within a functionally differentiated scientific system, where individual actors reside in the environment rather than constituting system elements. This contrasts with actor-network theory (ANT; ?), which foregrounds heterogeneous agents—authors, metrics, instruments—in assembling citation networks. While ANT richly traces local translations yet struggles to formalise emergent macro-structures, SSCT supplies the missing operational closure: once computed, our dual rankings reveal stable, self-referential hierarchies that condition future citations, structurally coupled to (yet autonomous from) the actor-networks ANT describes.

Spectral methods. Vigna [2016] provides an overview of spectral ranking methods. The lineage from Katz [1953] through Pinski–Narin [1976] to PageRank and the eigenfactor is traced by Franceschet [2011].

Duality of bipartite networks. A bipartite network has two sets of nodes connected only by edges between them. Spectral ranking over a bipartite network entails ‘duality’ [Breiger, 1974] whereby attention flows between the two node sets to settle stationary attention rankings over each. Our construction differs from Breiger’s duality in that all four blocks of the joint attention matrix are primary flows rather than derived projections.

Overview of the rest of the paper. Section 2 develops the mathematical framework in four layers: corpus construction (Layer 1), attention matrices (Layer 2), spectral ranking (Layer 3), and computation (Layer 4). Section 3 describes the economics and commerce corpus and reports descriptive statistics. Section 4 presents and interprets the rankings. Section 5 discusses validity and prospects.

2 Journal–institution citation networks

From reference to influence. The raw material of citation analysis is the reference: a bibliographic pointer written by a scientist and appended to a published work. Yet as Wouters [1999] argues, the citation as it appears in scientometric analysis is not identical to the reference produced at the scientist’s desk — it is the product of the citation indexer. The indexer resolves each reference string to a canonical work record, associating it with a unique identifier in the database, a process that involves author-name disambiguation, source-title normalisation, and deduplication of variant reference forms. Only after this resolution does a reference become a directed edge (w_i, w_j) in the citation graph, where both endpoints are identified works. The corpus construction below is a second, controlled layer of filtering built on top of the indexer’s first layer; the identification of references with works is presupposed, and its limitations for the specific database used in our empirical study are discussed in Section 3.

With references resolved to work-to-work edges, we introduce the analytical constructs that structure the rest of the paper. A *reference* from w_i to w_j transfers a unit of *attention* from w_i to w_j ; viewed from the receiving end, w_j receives a *citation* from w_i . Reference and citation are the same directed edge described from opposite ends. The unit of attention is not confined to a single pathway: it is distributed over source and institution channels using the weights established in the corpus construction, with the apportionment between the two channels governed by a parameter $\chi \in [0, 1]$ introduced in the next section. *Influence* is then the latent quantity that attention accumulates into recursively — each unit of attention received carries a weight proportional to the prestige of its source, which is itself determined by the influence that source has accumulated. Resolving this circularity is precisely the task of the spectral ranking.

Corpus construction. Let $\hat{\mathfrak{S}}$, $\hat{\mathfrak{W}}$, $\hat{\mathfrak{A}}$, and $\hat{\mathfrak{I}}$ denote the universal sets of sources, works, authors, and institutions respectively. The corpus is built by the following sequential pipeline, each step restricting the previous.

Step 1: source selection. Select a set of sources $\mathfrak{S} \subset \hat{\mathfrak{S}}$, $|\mathfrak{S}| = N_s$, by field classification (e.g. journal category lists from InCites or Scimago). The sources s_i , $i = 1, \dots, N_s$, define the disciplinary scope of the study.

Step 2: census works. Select the set of works $\mathfrak{W} \subset \hat{\mathfrak{W}}$, $|\mathfrak{W}| = N_w$, comprising all works w_i published in a source $s_j \in \mathfrak{S}$ within the *census window* $t_0^c \leq t_i \leq t_1^c$. These are the citing works of the study.

Step 3: target works. Each work $w_i \in \mathfrak{W}$ carries a reference list. A reference in that list to a work w_j is retained if w_j is published in a source $s_k \in \mathfrak{S}$ within the *target window* $t_0^t \leq t_j \leq t_1^t$. The set of all retained referenced works is $\mathfrak{W}^t \subset \mathfrak{W}$, $|\mathfrak{W}^t| = N_{w^t}$. The set of retained reference pairs is

$$\mathcal{R} = \{ (i, j) : w_i \in \mathfrak{W}, w_j \in \mathfrak{W}^t \}. \quad (1)$$

Step 4: induced authors and raw institutions. The census works induce a set of authors $\mathfrak{A}' \subset \hat{\mathfrak{A}}$ and, via their affiliations, a set of institutions $\mathfrak{I}' \subset \hat{\mathfrak{I}}$.

Step 5: institution filtering. Institutions with fewer than τ census works are excluded, yielding the ranked institution set $\mathfrak{I} \subset \mathfrak{I}'$, $|\mathfrak{I}| = N_u$. The threshold τ is a free parameter of the model (see Section 2).

Boundary flows and the open-corpus principle. The corpus boundaries generate four classes of flows that cross the boundary and are therefore only partially represented:

1. *Exogenous citing works.* Works outside $\mathfrak{W}$ (outside the census window or outside $\mathfrak{S}$) may reference works in $\mathfrak{W}^t$, contributing attention that the model does not observe.
2. *Out-of-scope references.* Census works $w_i \in \mathfrak{W}$ may reference works outside $\mathfrak{W}^t$ (outside the target window or outside $\mathfrak{S}$); these references are excluded from $\mathcal{R}$.
3. *Filtered institutions.* A census work w_i may be affiliated with institutions not in $\mathfrak{I}$; the institutional weight associated with those affiliations is lost.
4. *Source–institution allocation.* A single reference in $\mathcal{R}$ carries attention that must be apportioned between the source pathway and the institution pathway.

In cases (1)–(3) we adopt the *open-corpus principle*: flows that cross a boundary are excluded without renormalising the weights of the surviving flows. This treats the corpus as an open system in which external attention exists but is not modelled. The alternative — renormalising surviving weights to compensate for excluded flows — treats the corpus as closed and imposes a different, stronger assumption. We adopt the open-corpus principle as the baseline and note that the closed-corpus alternative is a robustness dimension within the parameter space θ (Section 2).

Work-level weight assignment. Each census work $w_i \in \mathfrak{W}$ is authored by A_i authors a_j , $j = 1, \dots, A_i$, where author a_j is affiliated with U_{ij} institutions from $\mathfrak{I}$. We assign to each institution $u_k \in \mathfrak{I}$ affiliated with w_i a weight ω_{ik} representing the fractional contribution of that institution to the work. Under *author-fractional* counting:

$$\omega_{ik} = \sum_{\substack{j=1 \\ k \in \mathcal{U}_{ij}}}^{A_i} \frac{1}{A_i} \cdot \frac{1}{U_{ij}}, \quad (2)$$

where $\mathcal{U}_{ij} \subseteq \mathfrak{I}$ is the set of retained institutions affiliated with author a_j of work w_i , and $U_{ij} = |\mathcal{U}_{ij}|$. Each author contributes a share $1/A_i$ of the work, and that share is divided equally among the author’s retained institutions.

Before institution filtering, $\sum_k \omega_{ik} = 1$ for every work w_i . After filtering, works whose affiliations include institutions not in $\mathfrak{I}$ will have $\sum_k \omega_{ik} < 1$; consistent with the open-corpus principle, these weights are not renormalised.

The same weight ω_{ik} applies to referenced works $w_j \in \mathfrak{W}^t$, using the authors and affiliations of w_j .

Model parameters. The corpus construction and weight assignment introduce the following free parameters, collected in the tuple

$$\theta = (\mathfrak{S}, [t_0^c, t_1^c], [t_0^t, t_1^t], \tau, \chi), \quad (3)$$

where $\mathfrak{S}$ is the source set, $[t_0^c, t_1^c]$ and $[t_0^t, t_1^t]$ are the census and target windows, $\tau \geq 1$ is the institution-size threshold, and $\chi \in [0, 1]$ is the source–institution allocation parameter introduced in the following section. Author-fractional counting is the baseline weighting scheme; institution-fractional counting (where $\omega_{ik} = 1/K_i$ for K_i distinct retained institutions of work w_i) is examined as a robustness check in Section 4. The sensitivity of rankings to variation across θ is examined systematically in Section 4.

From works to entities. The reference pairs $\mathcal{R}$ define a directed graph on $\mathfrak{W} \cup \mathfrak{W}^t$, with works as nodes. Each work carries two kinds of label: a source label $\sigma(i) \in \{1, \dots, N_s\}$ and a weighted set of institution labels $\{(k, \omega_{ik}) : k \in \mathcal{U}(i)\}$. We derive from this a weighted directed graph on the stacked node set of sources and institutions by aggregating the reference pairs over their labels — a weighted quotient of the work-level graph onto its label space [cf. Breiger, 1974]. The adjacency submatrices of this derived graph are the four blocks of the attention matrix defined below. No intermediate work-level matrix is required: the blocks are defined directly by summation over $\mathcal{R}$.

Attention pathways and the coupling parameter χ . A reference $(i, j) \in \mathcal{R}$ transfers one unit of attention from w_i to w_j . On the citing side, a fraction $(1 - \chi)$ of this unit is attributed to the source $s_{\sigma(i)}$ and a fraction χ is distributed over the institutions u_k , $k \in \mathcal{U}(i)$, proportionally to ω_{ik} . The same apportionment applies on the cited side. The parameter $\chi \in [0, 1]$ is the *source-institution coupling*: at $\chi = 0$ all attention flows through sources, recovering the conventional source-only setting; at $\chi = 1$ all attention flows through institutions; at $\chi = 1/2$ the two pathways carry equal weight. To our knowledge χ has no precedent in the bibliometric literature; it is the device that makes a joint source-institution ranking possible and that nests existing single-mode rankings as special cases.

The attention matrix $\mathbf{C}(\chi)$. Summing over all reference pairs $(i, j) \in \mathcal{R}$, with $p = \sigma(i)$ and $q = \sigma(j)$, yields the four blocks:

$$[C_{SS}]_{pq} = (1 - \chi)^2 \sum_{\substack{(i,j) \in \mathcal{R} \\ \sigma(i)=p, \sigma(j)=q}} 1, \quad (4)$$

$$[C_{SI}]_{pl} = (1 - \chi)\chi \sum_{\substack{(i,j) \in \mathcal{R} \\ \sigma(i)=p}} \omega_{jl}, \quad l = 1, \dots, N_u, \quad (5)$$

$$[C_{IS}]_{kq} = \chi(1 - \chi) \sum_{\substack{(i,j) \in \mathcal{R} \\ \sigma(j)=q}} \omega_{ik}, \quad k = 1, \dots, N_u, \quad (6)$$

$$[C_{II}]_{kl} = \chi^2 \sum_{(i,j) \in \mathcal{R}} \omega_{ik} \omega_{jl}, \quad k, l = 1, \dots, N_u, \quad (7)$$

assembled into

$$\mathbf{C}(\chi) = \begin{pmatrix} \mathbf{C}_{SS} & \mathbf{C}_{SI} \\ \mathbf{C}_{IS} & \mathbf{C}_{II} \end{pmatrix}, \quad (8)$$

of dimension $(N_s + N_u) \times (N_s + N_u)$, with rows indexing citing entities and columns indexing cited entities over the stacked node set.

All four blocks represent genuine attention flows generated by a single reference act: the source-to-source flow is as primary as the source-to-institution flow. This distinguishes $\mathbf{C}(\chi)$ from Breiger's 1974 duality, where the diagonal projections are derived objects rather than primary flows.

The column sums of each block of $\mathbf{C}(\chi)$ measure the raw weighted attention received by each entity, and provide a natural baseline for comparison with the spectral prestige scores derived in the next section.

From the attention matrix to the row-stochastic reference matrix. Let $r_p = \sum_q C_{pq}(\chi)$ be the total references given out by entity p under the attention matrix $\mathbf{C}(\chi)$, $\mathbf{D}_r = \text{diag}(r_1, \dots, r_N)$ with $N = N_s + N_u$, and $\mathbf{A} = \text{diag}(a_1, \dots, a_N)$ the diagonal matrix of work counts, where $a_p = N_p^s$ for source s_p and $a_p = M_k$ for institution u_k . We assume $r_p > 0$ for all p . The *row-stochastic reference matrix*

$$\mathbf{H}(\chi) = \mathbf{D}_r^{-1} \mathbf{C}(\chi), \quad (9)$$

has (p, q) entry $H_{pq} = C_{pq}/r_p$, the fraction of entity p 's references directed to entity q .

The Katz influence score. As established in the introduction, the Katz construction [Katz, 1953, Franceschet, 2011] assigns to each entity a score by summing contributions from all reference chains of all lengths terminating at it, with chains of length k discounted by α^k . A reference chain of length k reaching entity q passes through k intermediate referencing steps before arriving at q ; the attenuation α^k accumulates one factor of α at each such step, so that direct references ($k = 1$) contribute more than indirect ones. With damping parameter $\alpha \in (0, 1)$ and preference vector $\mathbf{p}$ satisfying $\|\mathbf{p}\|_1 = 1$, the Katz resolvent is

$$\boldsymbol{\pi}^*(\alpha) = (1 - \alpha) \sum_{k=0}^{\infty} \alpha^k (\mathbf{H}(\chi)^\top)^k \mathbf{p} = (1 - \alpha) (\mathbf{I} - \alpha \mathbf{H}(\chi)^\top)^{-1} \mathbf{p}, \quad (10)$$

which converges for all $\alpha < 1$ since the spectral radius of $\mathbf{H}(\chi)^\top$ equals one. As $\alpha \rightarrow 1$, $\boldsymbol{\pi}^*(\alpha)$ converges to the principal eigenvector of $\mathbf{H}(\chi)^\top$ — the stationary distribution $\boldsymbol{\pi}$ of the Markov chain with transition matrix $\mathbf{H}(\chi)$ — when that eigenvector is unique.

We set the preference vector to

$$p_p = \frac{a_p}{N_{\text{works}}}, \quad N_{\text{works}} = \sum_p a_p, \quad (11)$$

so that the prior weight of each entity is proportional to its work count: before reference chains are consulted, each published work contributes equally.

Following Geller [1978], π_q^* has a concrete interpretation: it is the long-run expected share of total reference flow received by entity q in the stationary behaviour of the reference system. Specifically, the expected number of references received by q per reference given out by q is

$$W_q = N_{\text{ref}} \frac{\pi_q^*}{r_q}, \quad (12)$$

where $N_{\text{ref}} = \sum_p r_p$ is the total reference flow in the system [Geller, 1978]. This is a concrete ratio of reference counts: references received per reference given, computed via the stationary structure of the reference network rather than by direct counting, and discounting indirect connections by the number of intermediary referencing steps separating them from q .

The *prestige per work* of entity p , in the sense of Palacios-Huerta and Volij [2004], is

$$\mathbf{v}^*(\alpha) = \mathbf{A}^{-1} \boldsymbol{\pi}^*(\alpha), \quad (13)$$

the long-run expected share of reference flow received per work published. An entity with $v_p^* = 2\bar{v}$ receives, per published work, twice the system-average share of reference flow. The score $\mathbf{v}^*$ can have equal elements: two entities with the same π_p^* and the same work count receive equal scores, and the preference vector (11) does not artificially break this symmetry. We treat α as a sensitivity parameter and report scores across a grid, typically $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$.

The fixed-point form of (10), $\boldsymbol{\pi}^* = \alpha \mathbf{H}(\chi)^\top \boldsymbol{\pi}^* + (1 - \alpha) \mathbf{p}$, is solved by successive substitution, converging geometrically at rate α since $\mathbf{H}(\chi)^\top$ is contractive on the probability simplex. The prestige per work (13) is then recovered as $\mathbf{v}^* = \mathbf{A}^{-1} \boldsymbol{\pi}^*$ once, after convergence; applying $\mathbf{A}^{-1}$ at each iteration instead would conflate reference-flow accumulation with work-count reweighting at every step — summing the averages rather than averaging the sum — yielding a fixed point with no clean interpretation.

Units and sparsity. Following Pinski and Narin [1976] we use the term *unit* for each node of the attention matrix $\mathbf{C}(\chi)$. Our unit set is $\mathcal{P} = \mathcal{S} \cup \mathcal{J}$, with $N = N_s + N_u$ units in total: the first N_s units are sources and the remaining N_u units are institutions, indexed $p = 1, \dots, N$. All quantities defined in Layers 1–3 — outgoing reference flow r_p , work count a_p , Katz score π_p^* , prestige per work v_p^* — are properties of units in this sense.

The matrix $\mathbf{C}(\chi)$ is of dimension $N \times N$ but is sparse: its non-zero entries correspond to unit pairs (p, q) for which at least one reference pair $(i, j) \in \mathcal{R}$ connects a work affiliated with unit p to a work affiliated with unit q . Since each work is published in exactly one source and affiliated with a small number of institutions, and since the reference pairs $\mathcal{R}$ are drawn from a bounded corpus, the number of non-zero entries satisfies

$$\text{nnz}(\mathbf{C}(\chi)) \leq 4|\mathcal{R}|, \quad (14)$$

one contribution per block per reference pair in the worst case. For the economics and commerce corpus, $|\mathcal{R}|$ is of order 10^6 while N^2 is of order 10^7 , so the matrix is substantially sparse and is never formed explicitly as a dense array.

Edge list and CSR construction. We construct $\mathbf{C}(\chi)$ from a list of weighted directed edges over units. Each reference pair $(i, j) \in \mathcal{R}$ contributes up to four edges, one per block of $\mathbf{C}(\chi)$, with weights determined by χ and the institutional weight vectors ω_{ik} :

$$\mathcal{E} = \{(\sigma(i), \sigma(j), (1-\chi)^2)\} \cup \{(\sigma(i), k, (1-\chi)\chi\omega_{jk}) : k \in \mathcal{U}(j)\} \cup \{(k, \sigma(j), \chi(1-\chi)\omega_{ik}) : k \in \mathcal{U}(i)\} \cup \{(k, l, \chi^2\omega_{ik}\omega_{jl})\} \quad (15)$$

where $\sigma(i)$ is the source index of work i and $\mathcal{U}(i)$ is the set of institution indices affiliated with work i . Edges with the same (citing unit, cited unit) pair are summed to give the entry $C_{pq}(\chi)$. The edge list $\mathcal{E}$ is then sorted by citing unit index p and converted to Compressed Sparse Row (CSR) format, in which the non-zero entries of each row p are stored contiguously in memory. Row normalisation by $r_p = \sum_q C_{pq}$ gives the CSR representation of $\mathbf{H}(\chi)$, and the transpose $\mathbf{H}(\chi)^\top$ required by the power iteration is formed without explicit transposition by traversing columns of the CSR structure. The dominant cost of each iteration is the sparse matrix–vector product $\mathbf{H}(\chi)^\top \boldsymbol{\pi}^{*(k-1)}$, which requires $O(\text{nnz}(\mathbf{C}))$ operations and is implemented directly in NUMPY/SCIPLY via `scipy.sparse.csr_matrix`.

Computation pipeline and parameter sensitivity. Figure 1 shows the full computation pipeline from the parameter tuple $\boldsymbol{\theta}$ to the prestige-per-work vector $\mathbf{v}^*$, annotated with the stage at which each parameter enters. Varying α alone requires only re-running the power iteration — the cheapest sensitivity test. Varying χ requires rebuilding $\mathbf{C}(\chi)$ from the fixed edge list $\mathcal{E}$ and re-running the iteration. Varying the institution threshold τ requires rebuilding $\mathcal{I}$, re-filtering $\mathcal{R}$, and reconstructing $\mathcal{E}$. Varying the time windows or source set $\mathfrak{S}$ requires re-querying the raw data and rebuilding the entire pipeline. The parameter grid explored in Section 4 is $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$, $\chi \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$, with τ and window choices examined as additional robustness dimensions.

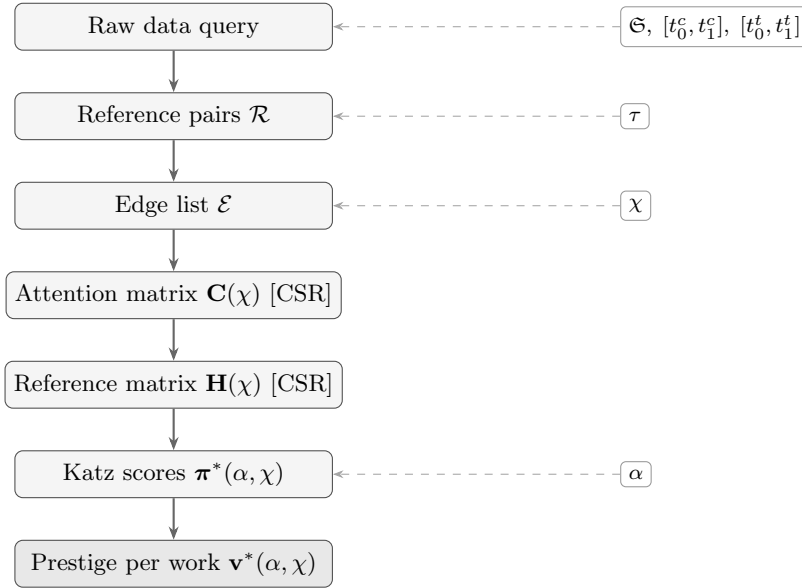


Figure 1: Computation pipeline from parameter tuple $\boldsymbol{\theta} = (\mathfrak{S}, [t_0^c, t_1^c], [t_0^t, t_1^t], \tau, \chi, \alpha)$ to prestige-per-work vector $\mathbf{v}^*$. Dashed arrows indicate the stage at which each parameter enters; rebuilding a stage requires rebuilding all stages below it.

3 Example: Economics and Commerce

Data sources and descriptive statistics. Guided by InCites EC journal categories, we extract OpenAlex data [Priem et al., 2022] for approximately 600 journals and 350,000 articles and reviews published between 2000 and 2025.

Network properties. [Placeholder.]

Limitations of the data. [Placeholder: reference cross-link, author disambiguation, institution identity.]

4 Results

Baseline model. [Placeholder.]

Exploring dependencies on model assumptions. [Placeholder.]

Parameter sensitivity. [Placeholder.]

Bootstraps. [Placeholder.]

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

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